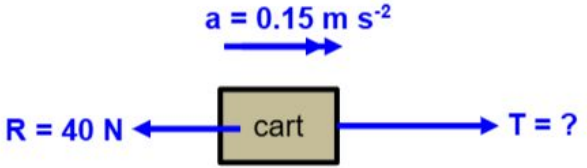
6	A lorry is pulling a cart of mass 600 kg with a rope at a constant speed along a straight horizontal road. The horizontal force resisting the motion of the cart is 40 N. The lorry now accelerates at 0.15 m s^{-2}. What is the tension in the rope?							
	A	65 N	**B**	100 N	**C**	130 N	**D**	195 N

L2 **Answer: C**

Free-body diagram of the cart during the acceleration: (consider horizontal forces)



Apply Newton's Second Law of Motion to the cart:

$F_{net} = m\,a$

$T - 40 = 600\ (0.15)$

$T = 130$ N